

WHAT GRADIENTS OVER REASONING MISS: PROMPT EDITS REWRITE THE REASONING

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ABSTRACT

GREATER lets a small language model optimize its own prompt with the gradient of the answer loss over its own reasoning, generated once and held fixed. Does this signal predict what prompt edits do? We make the objective exact for edits that rewrite, delete or insert whole instructions, by superposing candidate blocks behind attention gates, and test it with three 8–9B models on eleven pools of their own block edits, collected under a protocol fixed before their data. It ranks these edits in the *wrong* direction (negative in 8 of 11 pools, pooled Spearman -0.26), while fresh greedy accuracy on a few questions points the right way. An exact decomposition explains why: edits act through the reasoning the model writes, whereas the objective sees only the change of the answer read off a given reasoning, a term that on the models’ own samples averages at most 1.2 accuracy points per edit against mean absolute held-out effects of 2.1–10.9. Reweighting the model’s samples by exact likelihood ratios fails its preregistered test, as a coverage argument predicts: edits move the reasoning distribution beyond what the samples cover. Even a single-token substitution changes the greedy reasoning on 79–93% of questions (numerical re-rolls of the unchanged prompt: 58–85%). Edits must therefore be judged on fresh reasoning, where verification decides: in exploratory analyses, accepting the best candidate on 16 or fewer generations lowers held-out accuracy, and a noise-calibrated acceptance margin limits this harm. [TODO: end-to-end: structural search and official GREATER, gradient vs. random shortlist].

1 INTRODUCTION

Automatic prompt optimization is most useful for small, locally deployed models that cannot rely on a stronger model to critique their prompts (Zhang et al., 2024). GREATER (Das et al., 2025) lets such a model optimize its own prompt: the gradient of the answer loss with respect to one-hot prompt tokens, taken over the model’s reasoning generated once and then held *fixed*, shortlists token substitutions, and regenerated reasoning decides which one is kept.

This gradient measures how an edit changes the answer read off a given reasoning, not how it changes the reasoning. Yet prompts are structured documents (task, strategy, procedure, output contract), revised by rewriting, deleting or inserting whole instructions precisely to change how the model reasons. Does a gradient over fixed reasoning predict what such edits do? We make GREATER’s fixed-reasoning objective exact for block edits of any length, so that a failure of the signal cannot be a failure of a relaxation (§3), and test it on eleven pools of block edits that Llama-3-8B, Qwen3-8B and Gemma-2-9B propose for their own prompts, collected under a protocol fixed before their data (the hypotheses registered for these pools, a fork margin and an answer-conditioned repair, both failed).

The signal points the wrong way (§4). The fixed-reasoning objective correlates negatively with the held-out accuracy change in 8 of 11 pools (pooled Spearman -0.26 , sign-flip $p = 0.017$), while fresh greedy accuracy on 8 of the same training questions is positive in 9 ($+0.28$). Even a single-token substitution changes the greedy reasoning on 79–93% of questions (numerical re-rolls of the unchanged prompt: 58–85%); its mean accuracy does not differ detectably from that of the re-rolls (permutation $p = 0.07$ – 0.88 ; exploratory), although edits flip correctness on more questions in three of four pools.

Edits act on the reasoning, which the signal does not see (§5). An edit’s effect decomposes exactly into a distribution term, carried by the reasoning the model writes, and a read-off term, of which the objective is a soft version on one reasoning. On the models’ own samples the read-off term averages at most 1.2 accuracy points per edit against mean absolute held-out effects of 2.1–10.9.

The principled repair fails its preregistered test (§6). Reweighting the incumbent’s samples by exact likelihood ratios reaches a pooled Spearman of only +0.03 and a best-3 regret above a random shortlist’s, as coverage predicts: importance sampling needs about e^{KL} samples (Chatterjee & Diaconis, 2018); in 10 of 11 pools the median edit moves the distribution by more than $\log 16$ nats (in the direction the incumbent’s samples can measure), and without tempering the median effective sample size is 1.4–5.1 of 16.

Verification decides (§7; exploratory). On fresh reasoning, ranking quality rises monotonically with the generations per candidate. Accepting the best candidate on 16 or fewer generations lowers held-out accuracy, a winner’s curse. In a replay at a fixed budget of candidate reads, adaptive allocation reduces this loss without removing it, whereas accepting only gains above two standard errors cuts the loss of uniform verification at up to 12 questions per candidate from 1.1–1.9 to 0.2–0.6 points. In these single-round replays, verification and a calibrated acceptance rule pay off, and the fixed-reasoning shortlist does no better than a random one. [TODO: end-to-end: structural search and official GREATER, gradient vs. random shortlist; breadth].

2 SETTING: SELF-OPTIMIZATION WITH GRADIENTS OVER REASONING

Let f_θ be a frozen causal language model, P a prompt and (x, y) a question with its gold answer. Under P the model writes a reasoning $r \sim \pi_P(\cdot | x)$ (greedy decoding is the $\tau \rightarrow 0$ limit of temperature- τ sampling); an extractor e (“Therefore, the final answer is”) is appended and the answer is read greedily, with correctness $c_P(x, r) \in \{0, 1\}$ and gold-answer probability $q_P(y | x, r) = p_\theta(y | P(x) \oplus r \oplus e)$. A prompt should improve its accuracy $J(P) = \mathbb{E}_x \mathbb{E}_{r \sim \pi_P(\cdot | x)} c_P(x, r)$. GREATER (Das et al., 2025) lets a small model optimize its own prompt. For a training question it generates the greedy reasoning $\hat{r}$ under the incumbent prompt and scores a candidate P' by the answer loss on that *fixed* reasoning,

$$\ell(P'; x, y, \hat{r}) = -\log q_{P'}(y | x, \hat{r}), \quad (1)$$

which we call the *fixed-reasoning objective*. Its gradient with respect to one-hot prompt tokens ranks the model’s own top- k token substitutions; the shortlisted candidates are verified by regenerating the reasoning on all training questions, and the one with the fewest errors (plus a small perplexity penalty) is kept. The gradient decides which candidates are tried, fresh reasoning which is accepted; task and optimizer model coincide. Structural edits (rewrite a block, delete an instruction, insert a procedure) change the prompt’s length, so the one-hot relaxation is undefined.

3 EXACT STRUCTURAL EDITS IN ONE SEQUENCE

A prompt is a sequence of typed blocks $P = (b_1, \dots, b_m)$ (task, strategy, procedure, output, input, example); editable blocks may be empty insertion slots. The edits of slot j are deletion and a candidate set C_j proposed by the task model itself from unlabeled training questions (Appendix D). For one question, a single token sequence holds the incumbent prompt $P(x)$, every candidate of every slot, and a scored tail (the fixed reasoning and answer of Eq. 1, or a sampled reasoning). Candidate tokens of slot j attend, through the same gates, to everything before slot j (incumbent blocks and earlier slots’ candidates) and to their own earlier tokens; later tokens attend to the incumbent and to all candidates through gated attention

$$a_{tu} = \frac{G_{tu} \exp(s_{tu} - m_t)}{\sum_v G_{tv} \exp(s_{tv} - m_t)}, \quad G_{tu} = g_{\sigma(u)}, \quad (2)$$

where $\sigma(u)$ is the segment (incumbent block or candidate) of key u ; since the gate multiplies after the max-shift, $g = 0$ removes a key exactly. A continuous offset ω_j per slot moves every later token by $\Omega_t = \sum_j M_{tj} \omega_j$ positions ($M_{tj} = 1$ if token t follows slot j , else 0), as an extra rotary rotation of queries and keys.

Table 1: **Spearman correlation between each predictor and the held-out accuracy change of LLM-proposed block edits** (pooled: mean over pools, 95% bias-corrected paired bootstrap interval over questions, conditional on the pools). Answer: the fixed-reasoning objective; Fork: decision margin at reasoning forks (failed preregistered repair, Appendix L); each exact or patched (Eq. 5). Fresh n rows: fresh greedy accuracy on n training questions. LD3/LD7: logical deduction (3/7 objects); TS7/TS3: tracking shuffled objects (7/3; TS3 untouched). L/Q/G: Llama-3-8B/Qwen3-8B/Gemma-2-9B; *: non-initial prompt state; $\sqrt{\kappa}$: reliability bound of the target (Appendix D); cost: seconds per pool on one A6000.

Run	Answer exact	Answer patch	Fork exact	Fork patch	Fresh 8 rows	Fresh 24 rows	$\sqrt{\kappa}$
LD7-L	-0.55	-0.54	-0.14	+0.17	+0.48	+0.58	0.56
LD7-L*	-0.48	-0.59	-0.35	-0.23	-0.07	-0.15	0.76
LD7-Q	-0.25	-0.12	+0.18	+0.11	+0.18	+0.05	0.61
LD7-Q*	+0.24	+0.39	-0.21	-0.31	+0.34	+0.20	0.11
LD3-L	-0.41	-0.05	+0.02	-0.19	+0.27	+0.61	0.88
LD3-Q	-0.37	-0.30	+0.06	+0.02	+0.38	+0.42	0.84
TS7-L	-0.66	-0.42	-0.05	-0.13	+0.48	+0.57	0.95
TS7-Q	+0.03	+0.02	-0.58	-0.52	-0.30	+0.15	0.80
TS3-L	+0.08	-0.12	+0.11	+0.04	+0.36	+0.24	0.90
TS3-Q	-0.32	-0.31	-0.09	+0.12	+0.40	+0.23	0.90
TS3-G	-0.14	-0.17	+0.16	-0.41	+0.52	+0.70	0.93
Pooled	-0.26	-0.20	-0.08	-0.12	+0.28	+0.32	
95% int.	[-0.34,-0.24]	[-0.31,-0.14]	[-0.23,+0.04]	[-0.28,-0.02]	[+0.27,+0.37]	[+0.31,+0.41]	
Cost (s)	63	81	274	275	483	1064	

Proposition 1 (Exact vertices). *At gates 1 on incumbents, 0 on candidates and $\omega = 0$, the superposed computation equals $f_\theta(P(x) \oplus \text{tail})$. Choosing per slot one of {keep, delete, candidate c }, with the chosen candidate’s gate set to 1, the replaced incumbent’s to 0 and ω_j to the length change ($|c| - |b_j|$, $-|b_j|$, or $|c|$ for an insertion), it equals f_θ on the edited prompt, for any number of simultaneous edits of any lengths.*

The proof is by induction over layers (Appendix A, where Figure 3 contrasts the construction with GREATER’s token-level relaxation); fp32 tests confirm it for Llama, Gemma-2 and Qwen3 architectures, including combined edits (Appendix C). Because first-order gate derivatives saturate, a one-pass estimator, *renormalized patching*, recomputes downstream attention outputs with slot j swapped and linearizes only later layers (Eq. 5, Appendix A; the AtP* principle of Kramár et al., 2024): one forward and one backward pass score every single-slot edit.

4 DOES THE SIGNAL PREDICT WHAT STRUCTURAL EDITS DO?

Design. In each pool a model proposes 20–21 label-free rewrites and insertions for the slots of its incumbent prompt. Predictors, computed on 24 training questions, are the fixed-reasoning objective (exact and by one-pass patching) and, as reference, fresh greedy accuracy on 8 and on all 24 questions; the target is the change in greedy accuracy on 200 held-out questions. Eight pools replicate on Llama-3-8B and Qwen3-8B (three tasks, one non-initial prompt state) under a protocol fixed before their data for the fork hypothesis H-fork, which failed (Appendix L); three test an untouched task prospectively (TS3, three models). Pools are the units of inference (sign counts, exact sign-flip tests, made primary after these data were collected); bootstrap intervals are conditional on the pools. Best-3 regret—the held-out change of the three truly best edits minus that of a predictor’s top three—measures what a shortlist needs, against the exact expectation for a random shortlist; it replaces the registered best-5, under which the conclusions are unchanged.

The fixed-reasoning objective ranks block edits in the wrong direction. Computed exactly, it correlates negatively with the held-out change in 8 of 11 pools (pooled Spearman -0.26 ; sign-flip $p = 0.017$; Table 1); its one-pass estimate tracks the exact values (mean Spearman 0.79) and is negative in 9 of 11 (-0.20 ; $p = 0.037$). Fresh accuracy on the same questions is positive in 9

(8 questions; $+0.28$, $p = 0.010$) and 10 (24 questions; $+0.32$, $p = 0.004$) of the 11 pools. As a shortlist the patched objective is worse than a random one in 8 of 11 pools (best-3 regret 0.091 vs. 0.067; $p = 0.022$); the exact one is not distinguishable from random (0.075; worse in 7 of 11, $p = 0.14$), and on the untouched task its correlation is weaker (-0.13 , negative in two of three pools; patched -0.20 , negative in all three). The mean correlation is negative for every model and task, though its size varies, and it is negative on the two pools with little sample truncation as well (§10). On wrongly answered questions the negative correlation persists for Llama-3 but not for Qwen3 (Appendix E.3); two preregistered repairs failed (Appendix L).

Even a single-token edit changes most greedy reasoning. We re-read the token-level control (Llama-3 and Qwen3 on LD7 and TS7, 24 GREATER-style single-token substitutions per pool; Appendix E) with an engine whose same-prompt re-reads are token-identical (Appendix F). On average, one substitution changes the greedy reasoning on 79–93% of 200 held-out questions, first diverging after a median of 42.5–164 tokens, and flips correctness on 15–20% (Table 3, Appendix E.1). Reproducible reads are not batch-invariant (Yuan et al., 2025; He & Thinking Machines Lab, 2025): the unchanged incumbent read in requests of 16 to 100 questions changes the reasoning on 58–85% of questions and flips correctness on 10–16%. With these numerical re-rolls as the null (exploratory, specified after seeing the descriptive numbers), no pool shows a significant difference in mean accuracy between edits and re-rolls (exact permutation $p = 0.07$ – 0.88 ; pooled $p = 0.31$), although edits flip correctness on more questions in three of four pools. Greedy accuracy on 200 questions thus has a numerical floor of 1–2 points (sd), even with a deterministic engine; single-token edits sit at it, where no predictor is distinguishable from zero (Table 2), while block edits move held-out accuracy by 2.1–10.9 points on average.

5 WHAT A FIXED-REASONING GRADIENT CAN AND CANNOT SEE

An edit $P \rightarrow P'$ changes accuracy J through the reasoning the model writes and through how an answer is read off a given reasoning. With the likelihood ratio $w(r) = \pi_{P'}(r | x) / \pi_P(r | x)$ and $\mathbb{E}_{\pi_P}[w g] = \mathbb{E}_{\pi_{P'}}[g]$, exactly (for $\tau > 0$, where every reasoning has positive probability under both prompts),

$$J(P') - J(P) = \underbrace{\mathbb{E}_x \mathbb{E}_{r \sim \pi_P} [(w(r) - 1) c_{P'}(x, r)]}_D + \underbrace{\mathbb{E}_x \mathbb{E}_{r \sim \pi_P} [c_{P'}(x, r) - c_P(x, r)]}_A. \quad (3)$$

D : distribution term A : read-off term

The fixed-reasoning objective (Eq. 1) is a soft version of A on a single reasoning, the incumbent’s greedy one, and does not measure D . To first order, $D \approx \text{Cov}_{\pi_P}(\log w, c)$: an edit helps if it makes the model’s own *correct* reasoning more likely relative to its incorrect reasoning—the score-function term that a gradient over a fixed reasoning drops. If reading is faithful (answers read under P' and P differ only on reasonings of π_P -probability at most ε), then $J(P') - J(P) = D \pm \varepsilon$; on a question whose reasoning supports a wrong answer, every decrease of the objective raises the probability of reading an answer the reasoning does not support (Proposition 2, Appendix A).

The read-off term is small; the distribution moves far. On the incumbents’ own samples, the hard read of a fixed reasoning changes under a candidate in only 0.0–3.2% of (candidate, question, sample) triples across the eleven pools. The absolute read-off term averages 0.0–1.2 accuracy points per edit against mean absolute held-out effects of 2.1–10.9, at most about a third of the effect in any pool, and correlates with the held-out change at -0.31 to $+0.51$ across pools (pooled $+0.018$; Tables 4 and 6). The distribution, by contrast, moves far: $\text{KL}(\pi_P \parallel \pi_{P'}) = -\mathbb{E}_{\pi_P}[\log w]$ has a median of 1.1–14.6 nats per reasoning across pools (Table 4): the reasoning a fixed-reasoning gradient is taken over is rarely one the edited prompt would write.

The natural repair: reweight the model’s own samples. With GREATER’s white-box access, D can be estimated without new generation: draw S samples $r_s \sim \pi_P(\cdot | x)$ once per training question, read them under the incumbent, and compute the exact log-ratios $\log w_s$ of the *same* samples under every candidate. The tempered self-normalized importance sampling (SNIS) estimate

$$\hat{J}_\beta(P' | x) = \frac{\sum_s w_s^\beta c_P(x, r_s)}{\sum_s w_s^\beta}, \quad \hat{\Delta}_\beta(P') = \mathbb{E}_x [\hat{J}_\beta(P' | x) - \frac{1}{S} \sum_s c_P(x, r_s)] \quad (4)$$

estimates D (with the two reads exchanged), and the full change with candidate reads $c_{P'}$ in the numerator. At $\beta = 0$ it is the read-off term A on the incumbent's samples (candidate reads) or a sampled analogue of the fixed-reasoning objective (soft reads); $\beta = 1$ is consistent as $S \rightarrow \infty$; for small β , $\widehat{\Delta}_\beta \approx \beta \widehat{\text{Cov}}(\log w, c)$. Because weights over hundreds of tokens are heavy-tailed, β is chosen *without labels* as the largest value in $\{1, \frac{1}{2}, \frac{1}{4}\}$ whose median effective sample size $(\sum_s w_s^\beta)^2 / \sum_s w_s^{2\beta}$ is at least $S/4$. No reasoning is generated under any candidate.

Coverage predicts that reweighting fails. Importance sampling needs about $\exp(\text{KL}(\pi_{P'} \parallel \pi_P))$ samples (Chatterjee & Diaconis, 2018); incumbent samples give only the reverse divergence, so its comparison with $\log 16 \approx 2.8$ nats (exceeded by the median in 10 of 11 pools) is indicative, and the effective sample size at $\beta = 1$, 1.4–5.1 of 16, shows the shortfall directly. Tempering trades coverage for bias toward the fixed-reasoning family, and self-normalization hides the failure (Appendix G). The prediction is that no β yields a rank signal. It was written after five of the eleven pools had been observed (Appendix L); the registered H-dist test, fixed before any of its data, asks whether this principled gradient over the distribution of reasoning beats a random shortlist (§6).

6 REWEIGHTING THE MODEL'S OWN SAMPLES FAILS

On the eleven pools we drew $S = 16$ reasonings per training question under the incumbent ($\tau = 0.7$, no top- k /top- p truncation) and computed exact tempered likelihood ratios under every candidate (Appendix G). The primary predictor, registered for hypothesis H-dist, is Eq. 4 with candidate reads at the label-free β^* ($\frac{1}{2}$ in eight pools, $\frac{1}{4}$ in two, 1 in one). Its kill criterion was fixed before any of this data existed (Appendix L): beat a random shortlist in pooled best-3 regret (paired bootstrap $P(\text{gain} > 0) \geq 0.9$, and in at least 7 of 11 pools), and fall no more than 0.05 below fresh greedy accuracy on 8 questions (fresh-8) in pooled Spearman. H-inc applies it to the distribution term alone (incumbent reads); defined after the first pool, it is confirmatory on the other ten.

Both registered hypotheses fail all three conditions. The primary predictor has a pooled Spearman of +0.034 (bias-corrected interval $[-0.065, +0.190]$, conditional on the pools). Its best-3 regret, 0.075, is higher than the random shortlist's 0.067 (gain -0.009 $[-0.027, +0.008]$, $P(\text{gain} > 0) = 0.21$; lower regret in 6 of 11 pools), and its Spearman is 0.243 below fresh-8. On pools 2–11, H-inc has pooled Spearman -0.024 , regret 0.083 against 0.069 for random (6 of 10 pools, $P = 0.18$) and a Spearman 0.280 below fresh-8. H-dist and H-inc are not supported (Table 6, Appendix G).

No tempering recovers a signal, and the score is noise. Across $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, 1\}$ all estimates (candidate, incumbent or soft reads) have pooled Spearman -0.016 to $+0.060$, every bias-corrected interval containing zero (Figure 1); with soft reads at $\beta = 0$, the objective's sampled analogue is null (-0.016), against -0.258 on greedy reasoning. Regret falls toward $\beta = 1$ (0.056 – 0.062) without a rank signal, and the first-order score function is worse than random (regret 0.089). The primary score's split-half reliability is -0.31 to $+0.26$ per pool (a diagnostic added after five pools, like a KL gate that stays below fresh-8; Table 6, Appendix G).

7 VERIFICATION BUDGET DECIDES

What an edit does must therefore be observed on fresh reasoning, a noisy instrument even under a deterministic engine: if the fraction f of questions whose correctness flips between two prompts carries no systematic sign, their accuracy difference on n questions has variance f/n , a standard deviation of about 14–16 points on 8 questions and 3 on 200 at the flip rates of single-token edits (Appendix I). These analyses are exploratory; the matched-cost and fixed-budget replays were specified after the registered outcome, the pipeline simulation of Appendix H before it. An edit is scored by its mean correctness change over the incumbent on G generations per candidate, greedy (G questions) or sampled (k samples at $\tau = 0.7$ on each of r questions, $G = rk$; incumbent on the same seeds or on independent ones), averaged over 400 random subsets (Figure 2).

Quality rises with generations, not with sampling or coupling. Pooled Spearman rises monotonically with G for every verifier, from +0.178 for 4 greedy reads to +0.325 for 24 and +0.413

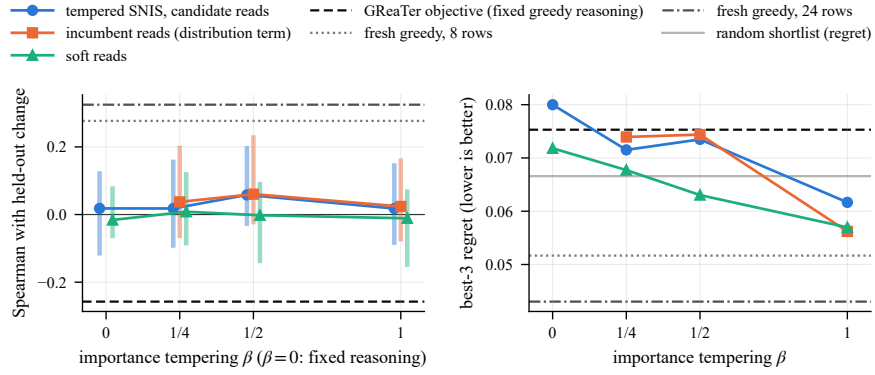


Figure 1: **Reweighting the incumbent’s samples does not recover a ranking signal at any tempering** (registered H-dist test, 11 pools; $\beta = 0$: no reweighting). Left: pooled Spearman with the held-out change for SNIS with candidate, incumbent (distribution term only) and soft reads; bars: bias-corrected bootstrap ranges. Lines: the fixed-reasoning objective on the greedy reasoning (GREATER objective) and fresh greedy accuracy on 8 and 24 training questions (rows). Right: best-3 regret (lower is better); random shortlist 0.067.

for 24×4 samples. At equal G , sampled reads rank worse than greedy ones in all 15 matched comparisons (by 0.03–0.06, none significant alone), and common random numbers (Corvelo Benz et al., 2025) change Spearman by at most 0.005 over independent seeds, as expected if edits rewrite the reasoning.

Small budgets lose accuracy. Accepting the best edit whenever its verified change is positive (verify-all) lowers held-out accuracy for every verifier with $G \leq 16$ (−0.54 to −1.71 points) and gains 0.41 points with 24 greedy reads (every sampled verifier at $G = 24$ still loses 0.21–0.58) and 1.44 at $G = 96$. With fixed questions and another tie rule (Appendix H), 8, 16 and 24 greedy reads realize −2.27, −0.77 and +1.14 points: the losses at small budgets are robust, the size of the gain at $G = 24$ is not.

The acceptance rule limits the loss more than the allocation does. A second replay fixes the budget B of candidate generations for one verification step (Appendix I, Table 7). Below the full budget, successive halving and a random shortlist of three lose less than uniform verification (0.68–1.57 and 0.60–1.05 points against 1.09–1.90), but neither gains at $B = 480$ (−0.24, −0.72; uniform +0.31), and racing loses at every budget; halving and racing follow budget-aware prompt selection (Shi et al., 2024; Schneider et al., 2025). Under uniform verification, accepting the best edit only if its paired gain exceeds two standard errors, $2\sqrt{f/n}$ for the pool’s flip rate f , keeps the loss at 0.24–0.62 points for $B \leq 240$ and raises the full-budget gain to +0.86 points. Whether the shortlist signal still matters under full verification is the question of §8.1.

8 SELF-OPTIMIZATION END TO END

8.1 END-TO-END STRUCTURAL SEARCH

End to end, task and optimizer model are GREATER’s Llama-3-8B-Instruct and Gemma-2-9B-it, on its 21 BIG-Bench Hard tasks (Suzgun et al., 2023) (50 train, 100 dev, up to 100 test questions), GSM8K (Cobbe et al., 2021) and FOLIO (Han et al., 2024), split with a fixed seed; 41% of our BBH test questions were used to produce GREATER’s published prompts, so comparisons with those prompts are also reported on the clean remainder. The structural search (Algorithm 1, Appendix J) runs on Llama-3-8B and seven tasks (six BBH tasks and FOLIO) for 12 rounds ($K = 6$ proposals per slot, shortlist 3), in arms that differ only in shortlist or verification: patch-v50 (fixed-reasoning objective) and random-v50 (random shortlist), both verified on all 50 training questions, and random-v8 (8-question minibatches); textgrad-v50 (textual gradients) is a different method class

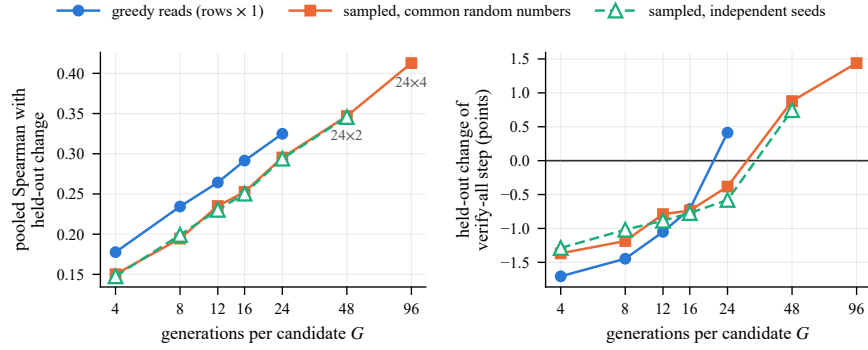


Figure 2: **Verification quality is set by the number of fresh generations** (exploratory; 11 pools, 400 random question/sample subsets per cell). G : generations per candidate; rows: training questions; sampled verifiers use one sample per question up to 24 questions, then 24×2 and 24×4 . Left: pooled Spearman with the held-out change. Right: held-out change realized by a verify-all step (ties broken uniformly).

with its own proposals. The distributional arm, registered conditional on H-dist or H-inc, is not run. The registered comparisons, paired by seed within tasks, are patch-v50 vs. random-v50 (does the shortlist signal matter under full verification?) and random-v50 vs. random-v8 (does the verification budget matter?); the prompt selected on dev is evaluated on test once, with three numerical re-rolls (deterministic engine, 4,096 tokens). [TODO: Seeds 1–3: test accuracy per task and mean (full, clean); paired differences, conditional and task-level intervals, W/T/L; textgrad-v50 (seeds 1–2); truncation; GPU time, tokens; the seven tasks.]

8.2 DOES GREATER NEED ITS GRADIENT?

The official GREATER code (Llama-3-8B, our 50 training questions, 106 steps, top- k 40, top- q 6, batch 64; memory-only patches, Appendix K) runs with its gradient shortlist and with a shortlist drawn uniformly from the same model-proposed candidates, on tasks fixed in advance because GREATER’s published prompts gain most there: tracking shuffled objects (five objects) and geometric shapes. [TODO: Gradient vs. random shortlist: test accuracy of final prompts (deterministic engine; full, clean) vs. the initial prompt; two registered repetitions per arm (candidate sampling unseeded); wall time.]

8.3 BREADTH

[TODO: Configuration chosen end to end (random-v50 if it matches patch-v50) on 21 BBH tasks, GSM8K and FOLIO, Llama-3-8B and Gemma-2-9B, vs. ZS-CoT and GREATER’s initial and published prompts; full and clean subsets; 4,096 tokens; truncation and cost.]

9 RELATED WORK

Gradient-based prompt optimization. AutoPrompt (Shin et al., 2020), GCG (Zou et al., 2023) and PEZ (Wen et al., 2023) rank token substitutions by task-loss gradients over relaxed prompt tokens; GREATER (Das et al., 2025) takes this gradient after the model’s own reasoning, so a small model can optimize its own prompt without a stronger critic. One-hot gradients poorly predict a token swap’s loss change (Li et al., 2024), evaluation dominates search cost (Zhao et al., 2024), and policy-gradient methods score sampled prompts by reward on fresh outputs (Deng et al., 2022; Diaio et al., 2023). Our exact objective still fails: the fixed reasoning, not the linearization, limits it.

LLM-proposed and structural prompt optimization. Most optimizers let a language model propose candidates and decide by measured scores or critiques: APE (Zhou et al., 2023), OPRO (Yang et al., 2024), ProTeGi (Pryzant et al., 2023), EvoPrompt (Guo et al., 2024), TextGrad (Yuksegonul

et al., 2025), MIPRO in DSPy (Opsahl-Ong et al., 2024; Khattab et al., 2024), PromptWizard (Agarwal et al., 2025) and GEPA (Agrawal et al., 2026). Textual gradients improve prompts without behaving like gradients (Melcer et al., 2025), random sampling is a strong baseline (Lu et al., 2024), and small models are weak optimizers in OPRO’s loop (Zhang et al., 2024). Structure-aware variants edit sections or typed units (SAMMO, SPRIG, SEPO, section-local textual gradients, segment-level optimization, prompt codebooks; Schnabel & Neville, 2024; Zhang et al., 2026; Ma et al., 2026; Sharma & Henley, 2026; Kulin et al., 2026; Nath et al., 2026), and GMPO, GradPO and PMPO locate segments by gradients or masking (Zhao et al., 2026; Chakma et al., 2026; Zhao et al., 2025). Most judge edits on fresh outputs, often on minibatches (Opsahl-Ong et al., 2024; Agrawal et al., 2026); we ask which selection signal a fixed proposer needs, and how much verification.

Evaluation noise and prompt sensitivity. Formatting changes move accuracy by tens of points (Sclar et al., 2024; Su et al., 2025); single-prompt evaluations rank models unreliably (Mizrahi et al., 2024), prompt optimization changes rankings (Sadjoli et al., 2025) and its gains shrink with scale (Zhou et al., 2025); greedy decoding varies with batch size, kernels and serving stack (Yuan et al., 2025; Attil et al., 2025; He & Thinking Machines Lab, 2025). Budget-aware evaluation casts prompt selection as best-arm identification or multi-fidelity search (Shi et al., 2024; Schneider et al., 2025), estimates performance from few evaluations (Polo et al., 2024) or selects informative subsets (Nian et al., 2026); p1 separates response noise from prompt-quality variance (Gao et al., 2026), and Gong & Wen (2026) ask which edits help. Here a single-token edit moves accuracy on average like a numerical re-roll, and an acceptance margin limits selection errors more than adaptive allocation does (exploratory).

Off-policy evaluation and importance sampling. Importance sampling, the classical tool of off-policy evaluation (Precup et al., 2000), needs about $\exp(KL)$ samples (Chatterjee & Diaconis, 2018); Pareto smoothing diagnoses heavy-tailed weights (Vehtari et al., 2024), and the effective sample size is a loose proxy for estimator quality (Elvira et al., 2022); coupled randomness reduces the samples needed to compare models (Corvelo Benz et al., 2025). Our exact likelihood ratios fail a preregistered test, as coverage predicts, and in exploratory replays common random numbers rank edits no better than independent seeds. Appendix B covers off-policy methods for language models, chain-of-thought faithfulness, attribution and block-wise contexts.

10 LIMITATIONS

Tasks, budget and scale. The pools come from two multiple-choice task families (logical deduction, tracking shuffled objects) and one proposal procedure; a closed answer set favours a stable read-off, which may be larger for free-form answers. At the validity budget of 384 tokens, incumbent samples are truncated on 46% (LD7-L), 97% (LD7-L*) and 100% (LD7-Q), and the incumbent’s greedy held-out reads on TS7-Q on 99.5% (Table 3), so these read-off terms, divergences and that target concern unfinished reasoning; sample truncation was not measured on the other six pools. On pools with little truncation (LD3-L 0.3%, TS7-L 14%) the objective is negative as well (-0.41 , -0.66). Models have 8–9B parameters and ran on one or two shared 48 GB GPUs: each pool is one proposal draw, Gemma-2-9B contributes one pool, the re-roll study covers two tasks and two models, and the verification analyses replay a single selection round, missing how selection errors compound, and set the acceptance margin from each pool’s flip rate over all of its reads rather than within the budget. The calculus and the likelihood ratios need white-box access; the calculus was verified on four rotary-embedding architectures (insertions and combined edits on three), the likelihoods against a direct per-token computation.

Engine and review. The re-roll null is specific to one deterministic engine configuration and one perturbation (request sizes of 16 to 100 questions); batch-invariant kernels (He & Thinking Machines Lab, 2025) would give a different, possibly degenerate, null, so token edits are shown to move accuracy as much as this engine’s numerics, not to be inert. The targets of Table 1 were read by a Hugging Face batched reader of unmeasured reproducibility, and the importance-sampling reads by an earlier engine configuration whose re-reads were not always token-identical (Appendix F); **[TODO: block re-reads of these targets under the deterministic engine, with re-rolls and truncation]**. The later analyses were reviewed only by fresh-context instances of the model family that ran the research, which cannot catch errors this family makes systematically.

11 CONCLUSION

Made exact for block edits, the fixed-reasoning signal of GREATER ranks LLM-proposed edits in the wrong direction, while fresh greedy accuracy points the right way. Edits act through the reasoning the model writes: the read-off term the signal sees is small, reweighting the model’s own samples failed its registered test as coverage predicts, and a single-token substitution changes the greedy reasoning on 79–93% of questions (numerical re-rolls of the unchanged prompt: 58–85%). Edits must be judged on fresh reasoning, where the verification budget decides: in exploratory replays, accepting the best candidate on 16 or fewer generations loses accuracy, and a two-standard-error margin limits the loss. [TODO: end-to-end: structural search, official GREATER gradient vs. random shortlist, breadth]. Building this rule into the search loop is the next step.

AI USE STATEMENT

Generative AI was used extensively in this work. An autonomous coding agent (Claude, Anthropic), directed by the authors, proposed and implemented the method, wrote the experiment and analysis code, ran the experiments and drafted the text. Adversarial reviews of the idea and of intermediate results were done by a model of a different family (GPT, OpenAI) in an early phase and later by fresh-context instances of the same model family that read the code and raw results directly. Related work was checked against the cited sources, and every numerical result is produced by scripts that read the recorded run outputs. The authors reviewed all AI-assisted work and take responsibility for the final content, including text, claims and artifacts.

REPRODUCIBILITY STATEMENT

Code for the superposed gated computation, the one-pass estimator, the distributional scores, the search loop, all baselines and all analyses is provided as supplementary material and will be released. Exactness propositions are proved in Appendix A and verified numerically (Appendix C); data splits are deterministic functions of the public GREATER release, and the released records list the example identifiers of every split, including the questions used to form the clean subset; the preregistered experiment plan with its dated amendments and all run records are included.

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A EXACT CALCULUS: ILLUSTRATION, PROOFS AND THE ONE-PASS ESTIMATOR

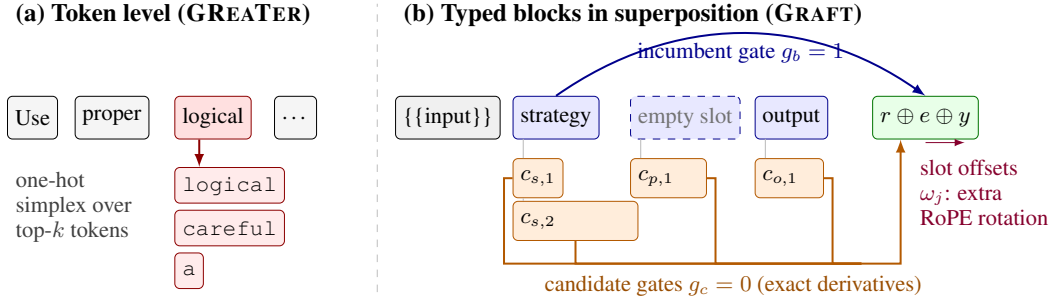


Figure 3: **The instrument.** (a) GREATER relaxes one prompt token over its top- k substitutions; vertices keep the prompt’s length and structure. (b) Candidate rewrites of each typed block sit in parallel slots behind attention gates, with slot offsets as rotary rotations; every vertex (replacement, deletion, insertion, combinations) is exactly an edited prompt (Proposition 1), and one forward and backward pass scores every single-slot edit (renormalized patching, Eq. 5). Both hold the incumbent’s reasoning r fixed; edits act mainly by changing r (§5).

Setting. We consider decoder-only transformers whose token mixing is causal softmax attention with rotary position embeddings, possibly with logit soft-capping and grouped-query attention; all other sublayers act on tokens independently. We require that edits *tokenize cleanly*: the edited prompt’s token sequence is the incumbent’s prefix tokens, the candidate’s tokens and the incumbent’s suffix tokens (edits violating this are excluded, which happens rarely because blocks end at line breaks). Sliding windows are assumed to exceed the sequence length, which our implementation asserts.

Proof of Proposition 1, base point. By induction over layers we show that every non-candidate token t has the hidden state of the reference computation on $P(x) \oplus r \oplus e \oplus y$. At the input, token identities and positions agree ($\Omega = 0$). At a layer, the keys visible to t are the non-candidate tokens that precede it in the reference order, with gate 1, and candidate tokens with gate 0. The latter

contribute $0 \cdot \exp(s_{tu} - m_t) = 0$ to both the numerator and the denominator of Eq. 2; the max-shift m_t may include them but softmax is shift-invariant. Keys and values of the former equal the reference ones by the induction hypothesis, and all other sublayers are tokenwise. $\square$

Proof of Proposition 1, edited prompts. Fix a choice $e_j \in \{\text{keep, delete, } c_j\}$ per slot and let P' be the edited prompt. Map every retained non-candidate token t and every chosen candidate token to its counterpart in P' . Its position in P' is its base position plus the cumulative length change of the edits before it, $\sum_{k \prec t} (|e_k| - |b_k|)$, which is exactly $\pi_t + \Omega_t$ at $\omega_k = |e_k| - |b_k|$ (candidates start at their slot’s first position and move only with earlier slots). The keys with non-zero gate visible to a mapped query are exactly the tokens preceding it in P' : removed incumbents and unchosen candidates have gate 0; a chosen candidate of slot j sees the prefix before slot j and the chosen candidates of earlier slots; a later token sees retained incumbents and chosen candidates. Rotary scores depend only on position differences, which agree with P' . The induction for the base point then applies verbatim to the mapped tokens, so the loss equals $\ell(P'; x, y, r)$. $\square$

Faithful reading. The statement summarized in §5:

Proposition 2 (Faithful reading). *If, for every candidate P' , the answer read under P' equals the answer read under P except on a set of reasonings of π_P -probability at most ε , then $|A| \leq \varepsilon$ and $J(P') - J(P) = D \pm \varepsilon$. Moreover, on a question whose reasoning r supports a wrong answer $a \neq y$, every decrease of the fixed-reasoning objective on r raises the probability of reading an answer that the reasoning does not support, and a decrease by $\delta > 0$ nats implies $q_{P'}(a | x, r) \leq 1 - e^\delta q_P(y | x, r)$.*

The proof is immediate from Eq. 3 and $q_{P'}(a | x, r) + q_{P'}(y | x, r) \leq 1$.

Saturation of gate derivatives. For one attention row, write the current output as $o_0 = N_0/Z_0$ with $N_0 = \sum_{u \notin c} G_{tu} e^{s_{tu}-m} v_u$, $Z_0 = \sum_{u \notin c} G_{tu} e^{s_{tu}-m}$, and let $S_c = \sum_{u \in c} e^{s_{tu}-m}$, $\bar{v}_c = S_c^{-1} \sum_{u \in c} e^{s_{tu}-m} v_u$. Then $o(g) = \frac{N_0 + g S_c \bar{v}_c}{Z_0 + g S_c}$, so

$$\left. \frac{\partial o}{\partial g} \right|_0 = \frac{S_c}{Z_0} (\bar{v}_c - o_0), \quad o(1) - o(0) = \frac{S_c}{Z_0 + S_c} (\bar{v}_c - o_0).$$

The first-order extrapolation exceeds the actual change by the factor $1 + S_c/Z_0$ in every head and layer, before any downstream nonlinearity. A candidate block that competes with an instruction block already holding most of a reading token’s attention has $S_c/Z_0 \gg 1$, so raw gate derivatives are dominated by attention mass rather than by the effect of the edit. The renormalized estimator below uses $o(1) - o(0)$ exactly.

Renormalized patching. Instead of gate derivatives, the one-pass estimator recomputes, from the exact keys and values present in the superposed pass, each downstream attention output with slot j swapped to an alternative a (query rotated by the length change), and linearizes only the propagation through later layers:

$$\widehat{\Delta \ell}(a) = \sum_{\text{layers } t \text{ after slot } j} \sum \langle \nabla_{o_t} \ell, o_t(a) - o_t(b_j) \rangle, \quad (5)$$

the AtP* principle (Kramár et al., 2024) applied to prompt edits. One forward and one backward pass score every single-slot edit of a sequence. Its fidelity against exact scoring is reported in §4 (mean Spearman 0.79 between patched and exact objectives over the eleven pools).

B RELATED WORK: FAITHFULNESS, ATTRIBUTION AND BLOCK-WISE CONTEXTS

Off-policy methods for language models. Off-policy prompt optimization has learned reward models from logged data (Sun et al., 2024) and smoothed weights over similar sentences (Kiyohara et al., 2025); Gumbel-max causal models generate counterfactual text (Ravfogel et al., 2025; Chatzi et al., 2025), and self-generated reasoning is a latent variable in training (Zelikman et al., 2022; Phan et al., 2023).

Faithfulness of chain-of-thought. Explanations may not reflect how a model answers (Turpin et al., 2023), answers depend on the reasoning to a varying degree (Lanham et al., 2023), and a few forking, critical or decision tokens can decide the outcome (Bigelow et al., 2025; Lin et al., 2024; Shen et al., 2026). The fixed-reasoning objective rewards unfaithful reads (Proposition 2), a channel too small to carry edit effects.

Attribution and block-wise contexts. Attribution patching and AtP* (Syed et al., 2024; Kramár et al., 2024), ContextCite (Cohen-Wang et al., 2024), AttriBoT (Liu et al., 2025) and AT2 (Cohen-Wang et al., 2025) estimate the effect of parts of a context; parallel context windows (Ratner et al., 2023), superposition prompting (Merth et al., 2024), Prompt Cache (Gim et al., 2024), Block-Attention (Ma et al., 2025), adaptive parallel encoding (Yang et al., 2025) and EPIC (Hu et al., 2025) encode or cache blocks independently with approximate positional handling. Our superposed sequence differs in purpose and guarantee: candidate blocks compete for the same slot behind gates, and continuous slot offsets make every vertex exactly the edited prompt, including length-changing and combined edits, which is what lets the fixed-reasoning objective be computed without approximation before it is tested.

C NUMERICAL VERIFICATION

All exactness claims are checked in fp32 on small randomly initialized models with a real tokenizer and chat template. The base point and every replacement and deletion vertex are checked on four architectures (Llama, Gemma-2 with attention and final logit soft-capping, OLMo-3 and Qwen3); insertion into an empty slot and a combined vertex with three simultaneous edits of different lengths on Llama, Gemma-2 and Qwen3. In all cases the superposed loss equals the loss of the stock model on the real edited prompt to within 10^{-4} nats. Gate and offset gradients agree with central finite differences to within $2 \cdot 10^{-3} + 2\%$. The weighted rationale objective, the decision margin and the tempered likelihood of a whole reasoning ($\tau = 0.7$; base point and a vertex) match stock computations to 10^{-4} , and renormalized patching reconstructs the base computation to 10^{-4} . A control confirms that approximate right- and left-padded layouts are *not* exact, so the tests detect the offset construction. The exact teacher-forced likelihoods of whole reasonings used for the distributional scores are checked against a direct per-token computation. The tests are part of the released code.

D EXPERIMENTAL DETAILS

Prompts and proposals. The initial prompt splits GREATER’s instruction into a strategy block (“Use proper logical reasoning and think step by step.”), an empty procedure slot and an output block (“Finally give the actual correct answer.”). The task model proposes candidates from two unlabeled training questions with a fixed meta-prompt per operator (rewrite: rephrase, specialize, add a step, shorten, warn about a pitfall, number the procedure, clarify the final answer; insert: procedure, verification, organization of facts, common mistake, comparison of options, restating facts, strategy hint, double-checking), sampled at temperature 0.8. Proposals are cleaned of preambles and type labels and must tokenize cleanly at block boundaries.

Reader. Greedy reasoning up to 1,024 tokens during search (384 in the validity studies, with the truncation rates of §10; 4,096 for dev selection and test), stopped at the chat template’s end-of-turn tokens; the extractor is appended inside the assistant turn (“Therefore, the final answer is (” for multiple choice; “(Yes or No) is:” style for binary tasks; “(a single integer) is:” for counting and arithmetic); 16 greedy answer tokens (8 in the v3 validity runs, which predate the reader fix) are parsed by answer type, tolerant of Markdown emphasis and $\LaTeX$ wrappers such as `\boxed{}`. Qwen3 runs in non-thinking mode. Rendered token ids are identical across the training-time and evaluation engines for all three models.

Validity studies. Per run: 2 training questions for proposals, 24 training questions for predictors, 6 sampled reasonings per question (temperature 0.7) for forks, at most 3 forks per question, and 200 held-out questions for the target; seeds 101 (v3), 202 with a non-initial state (v3), 303 (TS3). The validity tasks (logical deduction and tracking shuffled objects with three and seven objects) are BBH tasks outside GREATER’s 21; the benchmark of §8 contains the five-object variants of both families.

The exact objective takes one pass per edited prompt. The paired bootstrap resamples training and held-out questions jointly for all predictors (2,000 replicates) and keeps the proposal pools fixed, so intervals are conditional on the pools; they are bias-corrected in Table 1 and plain percentile intervals in Table 2. The split-half reliability κ of each held-out target bounds the attainable correlation, roughly by $\sqrt{\kappa}$ (Table 1). The cost row of Table 1 counts seconds per pool on one A6000, including the sampling that the fork predictors share.

Causal fork test. For 32 training questions per run, forks are computed as in the validity study. For each fork we force the correct-branch and the incorrect-branch token after the shared prefix and sample six continuations each; the control forces, at a uniformly random position of the shared prefix, the actual token versus the model’s most likely different token. Results are in Table 5.

E ADDITIONAL RESULTS

Table 2: **Token-level validity control:** single-token substitutions drawn from the model’s top-10 next tokens (t), same training and held-out questions as the block runs LD7/TS7 with seed 101. Columns as in Table 1, except that the intervals are plain percentile bootstrap intervals. The same edits are re-read under the deterministic engine in Table 3.

Run	Answer exact	Answer patch	Fork exact	Fork patch	Fresh 8 rows	Fresh 24 rows	$\sqrt{\kappa}$
LD7-L ^t	-0.09	+0.04	+0.00	+0.13	+0.15	+0.27	0.34
LD7-Q ^t	+0.23	+0.05	-0.11	-0.26	-0.05	-0.09	0.22
TS7-L ^t	-0.27	-0.29	-0.08	-0.13	+0.15	+0.17	0.31
TS7-Q ^t	-0.25	-0.45	-0.16	-0.13	-0.07	-0.10	0.51
Pooled	-0.10	-0.16	-0.09	-0.10	+0.04	+0.06	
95% int.	[-0.22,+0.11]	[-0.27,+0.05]	[-0.28,+0.10]	[-0.26,+0.12]	[-0.15,+0.19]	[-0.14,+0.20]	
Cost (s)	75	52	325	284	643	1254	

E.1 TOKEN EDITS AND NUMERICAL RE-ROLLS

Table 3: **Single-token edits versus numerical re-rolls of greedy decoding** (exploratory; 200 held-out questions, 384-token budget, deterministic engine). Edits: the 24 single-token substitutions per pool of the token-level control (Table 2). Re-rolls: the unchanged prompt read in separate requests of 16, 25, 40, 64 and 100 questions; the accuracy columns and tests also count the base read (one request of all 200 questions) as a re-roll. Reasoning changed, correctness flipped: mean share of questions whose greedy reasoning or correctness differs from the base read. sd: population standard deviation over reads. Permutation p : exact two-sided test over all $\binom{30}{6}$ relabelings of edits and re-rolls, for the difference of mean accuracies and the log ratio of accuracy variances; pooled: sum of per-pool standardized statistics, 2×10^5 joint relabelings. TS7-Q reads hit the budget on 99–100% of questions.

Pool	Reasoning changed (%)		Correctness flipped (%)		Accuracy (mean $\pm$ sd)		Permutation p	
	Edits	Re-rolls	Edits	Re-rolls	Edits	Re-rolls	Mean	Variance
LD7-L	87.5	69.1	20.3	14.4	0.462 $\pm$ 0.022	0.467 $\pm$ 0.018	0.65	0.72
TS7-L	83.7	57.5	19.0	9.7	0.319 $\pm$ 0.020	0.301 $\pm$ 0.022	0.07	0.47
LD7-Q	92.9	84.6	14.6	15.8	0.521 $\pm$ 0.020	0.517 $\pm$ 0.009	0.69	0.09
TS7-Q	79.2	67.7	16.1	11.7	0.273 $\pm$ 0.020	0.271 $\pm$ 0.009	0.88	0.16
Pooled							0.31	0.19

Details. Figure 4 shows every read of Table 3; the two same-prompt re-reads per pool are token-identical to the base read. The tests compare the 24 edits of a pool with six re-rolls (five chunked

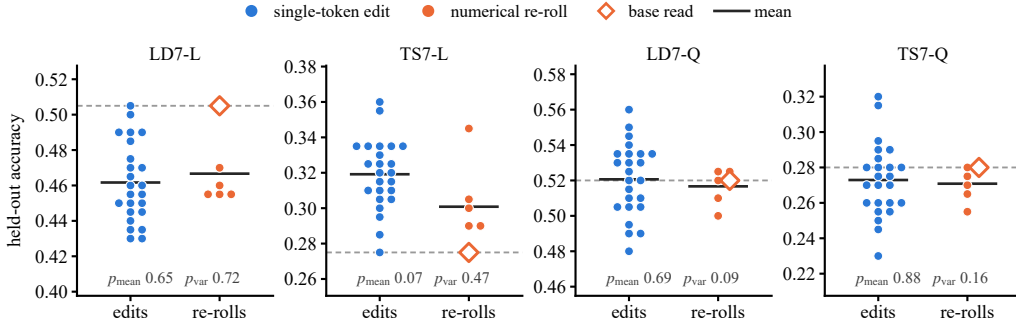


Figure 4: **Single-token edits and numerical re-rolls move held-out accuracy by similar amounts** (exploratory). Each dot is one read on 200 held-out questions: a single-token edit (blue) or the unchanged prompt read in requests of a different size (orange). Diamond and dashed line: the base read. Bars: means; the re-roll mean includes the base read. Every panel spans 13 accuracy points. p : exact two-sided permutation tests for the mean and the variance (Table 3).

reads and the base read). The spread test has little power with six re-rolls per pool: it does not detect a difference for Llama-3 ($p = 0.72, 0.47$), and for Qwen3, where edits spread about twice as wide as re-rolls, it reaches $p = 0.03$ only for the two Qwen3 pools combined, a subset formed after seeing the data. Edits flip correctness on more questions than re-rolls in three of the four pools (exact $p < 3 \times 10^{-4}$), and the additional flips cancel in accuracy.

A single greedy read is one draw. On LD7-L the base read lies 3.5 points above all five chunked re-rolls (0.505 vs. 0.455–0.470, sd 0.006), so 23 of 24 edits score below it. Leaving the base read out of the null (five re-rolls, all $\binom{29}{5}$ relabelings) gives mean $p = 0.86, 0.22, 0.67, 0.75$ (pooled 0.22) and variance $p = 0.03, 0.61, 0.18, 0.21$ (pooled 0.03) for LD7-L, TS7-L, LD7-Q and TS7-Q: LD7-L edits then spread more than the five tightly clustered re-rolls, and mean accuracies still agree. Edit accuracies spread close to or below what their pairwise disagreement implies if each disagreement’s sign were a fair coin (sample sd 0.022, 0.020, 0.020, 0.021 against 0.024, 0.024, 0.020, 0.021); the Qwen3 re-rolls are tighter than their disagreement implies (0.010 against 0.019 and 0.010 against 0.017), so the Qwen3 variance gap comes from the re-rolls rather than from unusually dispersed edits. First divergence is the position of the first differing reasoning token; when one reasoning is a prefix of the other, edits count the question at the shorter length and re-rolls leave it out, so the two groups’ medians are not directly comparable and we report them for edits only. Truncation was recorded only for TS7-Q, where the base and re-roll reads hit the 384-token budget on 99–100% of questions, so answers there are read off unfinished reasoning; it was not recorded for the other three pools, whose incumbents’ samples on the training questions are truncated at this budget on 46% (LD7-L), 14% (TS7-L) and 100% (LD7-Q).

E.2 READ-OFF TERM PER POOL

Table 4 gives the per-pool values summarized in §5: read-off flips of 0.0–3.2%, $|A|$ of 0.0–1.2 points against mean absolute held-out effects of 2.1–10.9, and median divergences of 1.1–14.6 nats per reasoning.

E.3 WRONGLY ANSWERED TRAINING QUESTIONS

Splitting training questions by whether the incumbent answers them correctly, and ranking each question’s loss changes across edits before averaging (which removes scale differences between questions), the negative correlation of the fixed-reasoning objective on wrongly answered questions persists for Llama-3 (−0.25 to −0.61 in all five pools) but not for Qwen3 (+0.25, −0.04, +0.07; its other two pools have one wrongly answered question each); Gemma-2’s pool gives −0.21. This is

Table 4: **The read-off term is small and uninformative** (Eq. 3; eleven pools; the incumbent’s 16 samples per training question, read under the incumbent and under each candidate). Read-off flips: fraction of (candidate, question, sample) triples whose hard read of the same reasoning changes; $|A|$: mean absolute read-off term per edit; $|\Delta_{\text{held-out}}|$, $|\Delta_{\text{fresh}}|$: mean absolute change of held-out greedy accuracy and of coupled fresh sampled accuracy on the training questions (accuracy points); ρ : Spearman between A and the held-out change; KL: median over edits of $-\mathbb{E}_{\pi_P} \log w$ per reasoning (nats); sd_x : mean over edits of the within-question standard deviation of $\log w$. Rows as in Table 1; TS3-G: $A = 0$ for every edit, so ρ is undefined.

Pool	Read-off flips	$ A $	$ \Delta_{\text{held-out}} $	$ \Delta_{\text{fresh}} $	$\rho(A, \Delta_{\text{held-out}})$	KL	$\text{sd}_x(\log w)$
LD7-L	1.4%	0.5	5.3	3.5	-0.31	10.2	4.6
LD7-L*	3.2%	1.2	3.4	3.8	+0.25	4.5	9.2
LD7-Q	1.4%	0.5	2.1	3.5	-0.19	3.9	3.9
LD7-Q*	0.9%	0.4	4.1	2.1	-0.03	1.1	4.8
LD3-L	0.2%	0.2	4.4	5.7	+0.28	10.6	5.4
LD3-Q	0.4%	0.2	2.4	2.8	-0.13	8.4	5.1
TS7-L	0.4%	0.2	7.7	6.4	-0.07	14.6	5.4
TS7-Q	2.0%	0.8	3.5	5.3	+0.51	4.9	3.4
TS3-L	0.3%	0.1	10.9	6.8	+0.02	8.9	5.4
TS3-Q	0.1%	0.1	3.8	2.3	-0.13	6.7	4.0
TS3-G	0.0%	0.0	8.8	10.7	–	10.4	6.2

consistent with the unfaithful-read mechanism of Proposition 2 for Llama-3; it does not show how much of the failure that mechanism accounts for.

E.4 FAILED REPAIRS AND AN EARLY PILOT

Table 5: **Causal fork test.** Mean change in success probability when forcing the correct instead of the incorrect fork token (ΔP_{fork}), its absolute value, and the absolute change at a random position of the shared prefix ($|\Delta P_{\text{ctrl}}|$).

Run	Forks	ΔP_{fork}	$ \Delta P_{\text{fork}} $	$ \Delta P_{\text{ctrl}} $
LD7-L	41	+0.085	0.199	0.191
LD3-L	41	+0.049	0.203	0.246
TS7-L	47	+0.092	0.220	0.234
LD7-Q	17	−0.069	0.186	0.144
LD3-Q	11	+0.136	0.258	0.146
TS7-Q	33	−0.025	0.157	0.181
Pooled difference $ \Delta P_{\text{fork}} - \Delta P_{\text{ctrl}} $		−0.008 [−0.041, +0.026]		

Rationale-likelihood objective (pilot). In an earlier pilot (validity v1; 8 training questions, 60 held-out questions; Llama-3 on date understanding and formal fallacies, Gemma-2 on date understanding), the likelihood of a self-generated gold-consistent rationale (PMPO-style) correlated 0.25, −0.16 and −0.02 with the held-out change, a correct-versus-incorrect rationale contrast 0.13, −0.22, −0.35, and a policy-gradient estimate −0.30 and −0.49; none was pursued. (The Gemma-2 pilot preceded a fix of its end-of-turn token in sampling and is reported for completeness only.)

Answer-conditioned shortlist (H-cond). Preregistered on the untouched task TS3 before its data existed: scoring edits by the answer-loss change only on training questions the incumbent answers correctly does not improve best-3 regret over the score on all questions (+0.000 [−0.025, +0.021]) and is worse than the exact expectation of a uniformly random ranking (−0.029 [−0.048, −0.002]). H-cond is retired.

F READING, DETERMINISM AND BUDGETS

Greedy decoding is not reproducible across batch compositions and cached computation paths (Yuan et al., 2025; He & Thinking Machines Lab, 2025). We found two sources in our generation engine (vLLM). With prefix caching, same-prompt re-reads changed 2–14% of answers, as large as the end-to-end effects under study. Without prefix caching but with the engine core in a separate process, the batch composition of the first decoding steps depends on when requests arrive: a first check (three reads, 200 questions) was token-identical, but with capped concurrency one of two re-reads changed the greedy reasoning on 55–66% of questions. With the engine core in the calling process, so that all of a call’s requests are enqueued before the first step, four re-reads of 200 questions were token-identical both at the evaluation configuration (4,096 tokens) and at the search memory configuration (384 tokens, a KV cache for about five 4,096-token sequences, so that sequences are preempted and recomputed). All end-to-end evaluations, re-reads and re-roll measurements use this configuration, with one prompt’s questions per request in a fixed order. The validity studies’ held-out targets (§4) were read by the Hugging Face model in the scoring process (batched greedy generation with left padding; re-read reproducibility not measured), and the H-dist samples and reads by the multiprocessing vLLM engine without prefix caching (Gemma-2: Hugging Face) [TODO: re-read the held-out targets of the eleven pools with the in-process engine and report the change of Table 1]. Rendered token ids are identical between the training-time (Hugging Face) and evaluation engines for all three models. Reasoning budgets are 1,024 tokens during search, 384 in the validity studies and 4,096 for dev selection and test; truncation rates are reported (for the validity studies in §10).

Numerical re-rolls. A reproducible engine is not batch-invariant. Reading the unchanged incumbent in separate requests of 16, 25, 40, 64 or 100 questions gives five reads that are each reproducible but differ from the single-request base read and from each other. In the four token pools the greedy reasoning differs from the base read on 67–71% (LD7-L), 56–61% (TS7-L), 83–86% (LD7-Q) and 65–73% (TS7-Q) of questions, correctness on 13–16%, 8–13%, 14–19% and 10–13%, and accuracy over the six reads has standard deviation 0.018, 0.022, 0.009 and 0.009. Batch-invariant kernels (He & Thinking Machines Lab, 2025) would make these reads identical; we keep the engine and use its re-rolls as the null against which edit effects are measured (§4, Table 3).

G DISTRIBUTIONAL SCORES: PROTOCOL

Per pool (the records of §4): $S = 16$ reasonings for each of the 24 training questions under the incumbent, sampled at $\tau = 0.7$ with no top- k /top- p truncation (so that the proposal is the full tempered softmax), the validity studies’ 384-token budget and a fixed seed per (question, sample). For every candidate and sample, the tempered log-likelihood of the reasoning is computed by teacher forcing in bfloat16 with float32 softmax, adding $\log P(\text{stop})$ at the end of samples that stopped. Hard reads (greedy, 16 tokens after the extractor) and soft reads (gold-answer probability) are recorded under the incumbent and under every candidate; four further samples per question under every prompt, with the same seed per (question, sample), give the coupled fresh evaluation. No estimator generates reasoning under a candidate. All estimators of Eq. 4 are computed afterwards from the stored values. Tempering trades coverage for bias toward the fixed-reasoning family at $\beta = 0$ (a smaller β targets the geometric mixture $\pi_P^{1-\beta} \pi_{P'}^\beta$), and self-normalization hides the failure of coverage: weights renormalized among samples that are all unlikely under the candidate look well-conditioned.

Registered predictors. Fixed before the data (Appendix L): the primary predictor (Eq. 4 with candidate reads at a label-free β^*), the β curve $\{0, \frac{1}{4}, \frac{1}{2}, 1\}$, soft reads, the first-order score-function estimate $\widehat{\text{Cov}}(\log w, c)$, effective-sample-size diagnostics, the coupled fresh evaluation, the held-out targets and the kill criterion of §6. The variant with incumbent reads (H-inc, the distribution term alone) was defined after the first pool and is confirmatory on pools 2–11. β^* is the largest $\beta \in \{1, \frac{1}{2}, \frac{1}{4}\}$ whose median per-question effective sample size $(\sum_s w_s^\beta)^2 / \sum_s w_s^{2\beta}$ over candidates is at least $S/4 = 4$ (else $\frac{1}{4}$). It selected $\beta^* = \frac{1}{2}$ in 8 pools, $\frac{1}{4}$ in 2 (TS7-L, TS3-G) and 1 in 1 (LD7-Q*); at $\beta = 1$ the median effective sample size was 1.4–5.1 of 16. The label-free rule thus met its effective-sample-size threshold in every pool, while in ten of the eleven pools the median divergence (3.9–14.6 nats per reasoning) exceeded $\log 16$, the regime in which the coverage argument predicts

failure with 16 samples. Verify-all for a predictor accepts the edit with the largest positive predicted change and records its held-out change (zero if none is positive).

Table 6: Registered H-dist test (11 pools): estimates from the incumbent’s own samples do not predict the held-out change; fresh reasoning does. Pooled Spearman: mean of the 11 per-pool Spearman correlations, with a 95% percentile bootstrap interval over pools (pools resampled with replacement, 20,000 replicates), wider than the bias-corrected question-bootstrap intervals of the text. Pools below random: best-3 regret strictly below the exact random expectation. Verify-all: accept the edit with the highest positive predicted change (first listed among ties); mean realized held-out change (points), pools improved/worsened (–: not simulated). GREATER objective: the fixed-reasoning objective (Eq. 1) on the greedy reasoning. Rows: training questions; sampled 24×4 : four samples on each of the 24 questions, with the incumbent read on the same seeds. Candidate reads, $\beta = 0$: the read-off term A on the incumbent’s samples. Primary: the registered predictor (β^* without labels). Incumbent reads: the distribution term alone, all 11 pools (H-inc is confirmatory on pools 2–11). KL gate: primary if $KL \leq \frac{1}{2} \log 16$, else fresh greedy on 8 questions; † added after five pools, before the Qwen3 and Gemma-2 data (a secondary analysis, Appendix L). Figure 1 plots the β curves.

	Pooled Spearman [pool bootstrap]	Best-3 regret	Pools below random	Verify-all (points)	Pools +/-
<i>Fixed reasoning</i>					
GREATER objective (exact)	-0.258 [-0.411, -0.095]	0.075	4/11	-5.00	+1/-10
Candidate reads, $\beta = 0$	+0.018 [-0.108, +0.156]	0.080	4/11	–	–
<i>Reweighted own samples</i>					
SNIS, $\beta = 1/4$	+0.018 [-0.193, +0.222]	0.072	6/11	–	–
SNIS, $\beta = 1/2$	+0.058 [-0.146, +0.252]	0.073	6/11	–	–
SNIS, $\beta = 1$	+0.018 [-0.177, +0.213]	0.062	6/11	-2.77	+4/-7
Primary, β^*	+0.034 [-0.185, +0.241]	0.075	6/11	-2.18	+5/-6
Incumbent reads, β^*	+0.023 [-0.212, +0.246]	0.076	7/11	-2.14	+6/-5
First-order score function	-0.098 [-0.285, +0.066]	0.089	3/11	-8.45	+2/-9
KL gate †	+0.218 [+0.030, +0.382]	0.052	8/11	–	–
<i>Fresh reasoning</i>					
Greedy, 8 rows	+0.277 [+0.122, +0.404]	0.052	9/11	-2.27	+3/-4
Greedy, 24 rows	+0.325 [+0.173, +0.473]	0.043	8/11	+1.14	+5/-4
Sampled, 24×4 (coupled)	+0.413 [+0.242, +0.572]	0.035	9/11	+1.00	+5/-5
Random shortlist	0	0.067	–	–	–

Further results. As an acceptance rule, SNIS at $\beta = 1$ loses 2.77 held-out points on average (4 pools improved, 7 worsened; Table 6). The first-order score function, the small- β limit, is worse than random (best-3 regret 0.089; gain $-0.023 [-0.041, -0.006]$). The primary score’s agreement with the on-policy estimate of the same quantity from fresh samples is -0.52 to $+0.36$ per pool. Coupled sampled accuracy on 24 questions $\times$ 4 samples has regret 0.035 (gain over random $+0.031 [+0.026, +0.045]$, 9 of 11 pools).

Diagnostics added after five pools. After the five Llama-3 pools and before any Qwen3 or Gemma-2 data, descriptive diagnostics were added for all eleven pools: split-half reliability of the primary score (samples 1–8 against 9–16), its agreement with the coupled fresh estimate, the on-policy read-off term (Table 4), the divergence $-\mathbb{E}_{\pi_P} \log w$ per reasoning, and a KL gate fixed from theory rather than data: an edit is scored off-policy only if this divergence is at most $\frac{1}{2} \log S = 1.39$ nats, and otherwise by fresh greedy accuracy on 8 questions. The gate keeps 0–52% of a pool’s edits off-policy (median 10%) and reaches a pooled Spearman of $+0.218$, below fresh-8 alone ($+0.277$). A numerical null re-scores the incumbent’s samples with a different batch size; the standard deviation of the log-ratio between the two passes measures bfloat16 and padding noise. On the six pools sampled after the amendment it is 0.4–1.3 nats per reasoning, against within-question standard deviations of $\log w$ of 3.4–6.2 nats on the same pools.

Intervals. Bias-corrected bootstrap intervals resample training questions and held-out questions within each pool and are conditional on the pools. For the coupled fresh evaluation no replicate fell below the point estimate, so its interval $([+0.415, +0.499])$ excludes the estimate $(+0.413)$.

H PIPELINE SIMULATION

For each pool and a verification size $n \in \{8, 16, 24\}$, one round of the shortlist-then-verify loop is replayed from recorded per-question correctness: the edits are scored by a predictor on the 24 training questions, the top three (or all edits) are verified by fresh greedy accuracy on the first n training questions, and the best is accepted if it beats the incumbent on those questions; the realized gain is the accepted edit’s change on the 200 held-out questions (zero if none is accepted). Ties among candidates go to the first-listed edit, as in the verify-all column of Table 6. The random shortlist is scored by its exact expectation over all three-subsets. With a shortlist of three verified on 8 questions, the answer-loss shortlist realizes -3.27 (exact) and -3.23 (patched) points and a random shortlist -1.34 ; verifying all edits on 8, 16 and 24 questions realizes -2.27 , -0.77 and $+1.14$; the best edit of each pool would add 5.50.

I VERIFICATION AT MATCHED COST: PROTOCOL

Noise of a paired comparison. Fresh reasoning measures an edit’s whole effect, but greedy accuracy on n questions is a noisy instrument even under a deterministic engine: numerical re-rolls of the unchanged incumbent rewrite much of its greedy reasoning and move its accuracy (§4, Table 3). Counting flips gives the size of this noise: comparing two prompts on n questions, the accuracy difference is $(g - h)/n$ for g gained and h lost questions, and if the $f = (g + h)/n$ flipped questions carry no systematic sign its variance is f/n . From the measured disagreement between single-token edits this predicts a standard deviation of 2.0–2.4 points for their accuracies (observed: 2.0–2.2). Single-token edits flip 15–20% of questions (Table 3), so comparing an edit with the incumbent has a standard deviation of about 14–16 points on 8 questions, 5–6 on 50 and 3 on 200, against mean block-edit effects of 2.1–10.9 points. Selecting the best of many candidates on such comparisons is a winner’s curse: the largest verified gain is mostly noise, and accepting it lowers held-out accuracy (§7).

Protocol. This analysis was added after the registered H-dist outcome and is exploratory. It uses the eleven pools’ recorded per-question correctness on the 24 training questions: greedy reads of every edit and of the incumbent, and four sampled reasonings per question and prompt at $\tau = 0.7$ that share one seed per (question, sample) across prompts (Appendix G). A verifier with G generations per candidate predicts an edit’s effect as the mean over questions of candidate minus incumbent correctness, in one of three forms: *greedy* reads on r questions ($G = r$, $r \in \{4, 8, 12, 16, 24\}$); *coupled* sampled reads, k samples on each of r questions with the incumbent read on the same k seeds (common random numbers; $G = rk$); and *independent* sampled reads, as coupled but with the incumbent read on k other seeds of the four ($k \leq 2$), so that candidate and incumbent reads are independent. The grid covers $rk \in \{4, 8, 12, 16, 24, 48, 96\}$ with $k \in \{1, 2, 4\}$ and $r \leq 24$; G counts the candidate’s generations, since the incumbent’s reads are shared by all candidates. For every cell and pool, 400 random draws of the r questions (without replacement) and of the k samples give the expected Spearman correlation with the held-out change, the best-3 regret and the realized held-out change of a verify-all step that accepts the edit with the largest predicted change if it is positive (ties in the prediction broken uniformly in both).

Comparisons. Pools are the units: coupled against greedy and coupled against independent verification at equal G are compared with exact sign-flip tests over pools. Greedy verification has the higher mean Spearman at every equal G , and no coupled-versus-greedy or coupled-versus-independent difference in Spearman reaches $p < 0.05$ (smallest $p = 0.052$). Figure 2 plots the cells with one sample per question and the 24×2 and 24×4 cells; mixed layouts fall in between (coupled 12×2 : $+0.281$). Because this analysis averages over question subsets while the pipeline simulation of Appendix H verifies on the first n questions, verify-all on 8 greedy questions realizes -1.45 points here and -2.27 there (§7).

Allocation and acceptance at a fixed budget. A second replay, specified after the results above and equally exploratory, uses the greedy reads of each pool’s $K = 20$ or 21 edits and of the incumbent on the 24 training questions. The budget B counts candidate reads of one question; the incumbent’s reads are shared and not counted. For every pool, budget and allocator, 400 random orders of the 24 questions are drawn and the allocator reads questions in that order. Unless a margin rule applies, it accepts its best candidate if the candidate’s mean paired correctness difference from the incumbent, on the questions it was read on, is positive (ties among candidates broken uniformly). The score is the accepted edit’s held-out change (zero if none), averaged over orders and then over pools; accepting the best edit of every pool would gain 5.50 points, and the mean edit changes held-out accuracy by -2.93 . *Uniform* verification reads all K edits on the first $\min(24, \lfloor B/K \rfloor)$ questions. The *random shortlist* reads three uniformly drawn edits on $\min(24, \lfloor B/3 \rfloor)$ questions, the pattern of Algorithm 1 with a random shortlist. *Successive halving* reads all edits on $\max(1, \lfloor B/(K \lceil \log_2 K \rceil) \rfloor)$ questions, keeps the better half by paired mean, doubles the number of questions (reusing those already read) and repeats while the budget allows. *Racing* adds one question at a time to all surviving edits and, after n questions, drops every edit whose paired mean lies more than $\sqrt{2f/n}$ below the leader’s, until one edit remains, the budget is spent or all questions are read. Here f is the pool’s flip rate, the fraction of (edit, question) pairs whose correctness differs from the incumbent’s (0.06–0.41 across pools); it is computed once per pool from all recorded reads, where a verifier would estimate it from earlier rounds. The *margin* rules verify uniformly but accept only if the paired gain exceeds $c\sqrt{f/n}$ with $c = 1$ or 2, the standard error of a mean of n paired differences of which a fraction f is non-zero (see the noise paragraph above); at 24 questions the 2-SE margin is 10–26 accuracy points.

Table 7: **Held-out change (points) realized by one verification step at a fixed budget** (exploratory; 11 pools, 400 question orders each). B : candidate reads of one question, the incumbent’s reads not counted. Rows per edit: training questions read per edit, $\lfloor B/K \rfloor$ for $K = 20$ –21 edits, at most 24. Margin rules: uniform verification, accepting only if the paired gain exceeds 1 or 2 standard errors $\sqrt{f/n}$. Best edit of each pool: $+5.50$; mean edit: -2.93 .

Budget B	40	60	80	120	160	240	480
Rows per edit (uniform)	1–2	2–3	3–4	5–6	7–8	11–12	22–24
Uniform	-1.70	-1.90	-1.60	-1.66	-1.39	-1.09	+0.31
Random shortlist of 3	-1.05	-0.75	-0.75	-0.60	-0.79	-0.78	-0.72
Successive halving	-1.57	-1.07	-0.99	-1.01	-1.03	-0.68	-0.24
Racing	-1.83	-1.94	-1.68	-1.52	-1.50	-1.24	-1.09
Uniform, gain > 1 SE	-1.73	-1.49	-0.87	-1.04	-1.13	-0.96	-0.07
Uniform, gain > 2 SE	-0.45	-0.62	-0.35	-0.38	-0.24	-0.43	+0.86

Results. Below the full budget every allocator loses held-out accuracy (Table 7). Successive halving and the random shortlist, which compare fewer candidates on more questions each, lose less than uniform verification there, but at $B = 480$ only uniform verification gains among the four allocators without a margin ($+0.31$; halving -0.24 , shortlist -0.72 , racing -1.09). This uniform value differs from verify-all on 24 greedy reads ($+0.41$, Figure 2) because the two pools with 21 edits read 22 questions and ties are drawn anew. The 2-SE margin has the smallest loss at every budget below 480, losing less than uniform verification in 6 or 7 of the 11 pools at each of these budgets, and the largest gain at $B = 480$. There every edit is read on all or nearly all questions, so each pool contributes essentially one decision: the margin leaves seven pools unchanged, in which uniform verification gained in two and lost in three, and one pool contributes $+10.5$ points under both rules. A 1-SE margin is too small: it loses 0.87–1.73 points below the full budget and 0.07 at it.

J STRUCTURAL SEARCH: ALGORITHM AND READING

Algorithm 1 lifts GREATER’s loop to blocks: the model proposes candidates, a signal shortlists them, and fresh reasoning on all training questions decides. The model optimizes its own prompt, and supervision enters only through the training labels: correctness, and the gold answer’s loss in the patch shortlist and in tie-breaks. The arms of §8.1 isolate the shortlist signal: the fixed-reasoning objective estimated in one pass (Eq. 5), or a uniformly random shortlist drawn from its

Algorithm 1 Structural shortlist-then-verify search (one round); patch-v50 and random-v50 differ only in line 3

Require: model f_θ , typed incumbent P , training questions D ($|D| = 50$), shortlist size μ

- 1: propose K label-free candidates per slot with f_θ from unlabeled questions; keep edits that tokenize cleanly at block boundaries and preserve required literals
- 2: compute the incumbent’s greedy reasoning on D
- 3: shortlist μ edits by the fixed-reasoning objective lifted to blocks (one pass, §3) or uniformly at random
- 4: regenerate greedy reasoning on all of D for the shortlisted edits
- 5: accept the best if it makes fewer errors on D than P (ties: answer loss)

own random stream, so that minibatches and proposals are identical across arms for a seed. Two controls lie outside this comparison. A textual-gradient arm, in which the same model critiques failed training questions and rewrites each block, is a different method class, not a matched-pool control; a verification-noise control verifies on random 8-question minibatches instead of all 50 questions. Dev questions select among the incumbents at four evenly spaced rounds; the selected prompt is evaluated on test once (no selection on test), with three numerical re-rolls at request sizes 20, 34 and 50.

Reading. One engine reads every prompt of a comparison with identical token-level rendering: greedy reasoning, the task-typed extractor inside the assistant turn and 16 greedy answer tokens parsed by answer type (Appendix D). Repeated reads are token-identical. The engine runs without prefix caching and with its core in the calling process, so every request of a call is enqueued before the first decoding step and batch composition does not depend on request timing; four re-reads of one prompt on 200 questions are identical at the evaluation and at the search memory configuration (Appendix F). Reproducible reads are not batch-invariant (§4), so every dev and test request holds exactly one prompt’s questions in a fixed order. During search the three shortlisted candidates share one generation call and the training questions are reshuffled each round, so a verification read there repeats only for an identical request. Reasoning budgets are 1,024 tokens during search and 4,096 for dev selection and test; truncation rates are reported.

K OFFICIAL GREATER ON ONE GPU

We run the official GREATER code with Llama-3-8B on our 50 training questions of each task, copied verbatim from GREATER’s own data files in our seeded order, with 106 steps, top- k 40, top- q 6 and batch 64. The released configuration loads the model twice, for a gradient worker and for a reasoning worker; we have one 48 GB GPU. Our patches change only where tensors live and when they are freed: (i) one local copy of the model serves both workers; (ii) the reasoning worker regenerates the training questions in calls of one eighth of them instead of one half, and every question is still evaluated; (iii) candidate logits (computed in GREATER’s default batches of 9) are freed before the next candidate is evaluated; (iv) the gradient worker’s weights are frozen and its CUDA cache is emptied after each task. Freezing removes about 16 GB of weight gradients that GREATER’s backward pass computed but never read; the gradient with respect to the one-hot prompt tokens, the only one used, is unchanged. The optimization algorithm, its hyperparameters, the evaluation of candidates by fresh reasoning and the selection rule are unchanged; smaller generation calls change batch shapes, which perturbs greedy decoding numerically as any change of batch size does (Appendix F). A two-step smoke run took 31 minutes with a peak of 33 GB. The patches are released with our code.

Shortlist ablation. One environment switch replaces the gradient ranking and nothing else: instead of the q candidates with the largest gradient score among the model’s k proposed tokens, the shortlist is q of the same k proposals drawn uniformly at random, and the gradient is not computed. Regeneration, scoring by fresh reasoning and selection are identical in both arms. The ablation was registered before its runs (Appendix L). [TODO: runs: tracking shuffled objects (five objects) and geometric shapes, gradient and random shortlist; test accuracy under the deterministic engine, full and clean subsets; wall time.]

L PREREGISTRATION AND AMENDMENTS

The experiment plan was fixed before any optimizer result and amended with dates and reasons (Stage C is the end-to-end structural search of §8.1): dev selection restricted to checkpoint incumbents; Stage C deferred until the validity study; H-fork and its replication protocol frozen after one exploratory result; evaluation separated from search with one shared engine; a reader fix (L^AT_EX-wrapped answers, answer length and reasoning budget) before any baseline or Stage C result; the Stage C variant set and its later re-scoping to 140 runs before any Stage C outcome; H-cond before its data; and the official GREATER shortlist ablation before its runs. After an internal review found the generation engine nondeterministic (prefix caching) and a truncation artifact, the reading protocol was fixed for every later evaluation read (no prefix caching, one prompt’s questions per request, a 4,096-token budget for re-reads, dev selection and test), proposals cut at their token cap were discarded, pool-level tests became primary, and the distributional hypothesis H-dist with its kill criterion was fixed before any of its data; its incumbent-read variant H-inc was defined after the first of eleven pools and is tested on the other ten. The Stage C protocol was then revised to verify on all 50 training questions (as GREATER does) before any of its runs. After five of the eleven H-dist pools, before any Qwen3 or Gemma-2 data, descriptive diagnostics, a KL gate fixed from theory and a numerical null were added (Appendix G), and the coverage prediction of §5 was first written down. When a second source of nondeterminism appeared (request timing with a multiprocess engine core), all evaluation, re-read and re-roll jobs and the Stage C search moved to the in-process engine, and the one Stage C run made before the change was archived and re-run. After the registered H-dist and H-inc outcomes, four analyses were added and are labelled exploratory: verification at matched generation budgets, allocation and acceptance at a fixed budget (both Appendix I; the latter is not logged in the plan), numerical re-rolls as the null for edit effects (§4), and truncation of held-out reads and of the H-dist samples. The pool-level sign-flip tests of the validity study (§4) were made primary after all eleven validity pools existed. The failed hypotheses (H-fork, H-cond, H-dist, H-inc) are reported as registered. Deviations: the v3 replication runs on a non-initial prompt state share their 200 held-out questions with the initial-state run of the same task and model (the protocol asked for independent slices), and the bootstrap does not resample pools; under the stricter preregistered causal statistic (signed $\Delta P_{\text{fork}} - |\Delta P_{\text{ctrl}}| = -0.156 [-0.203, -0.112]$) H-fork fails as under the lenient one. Table 1 reports best-3 regret, registered for TS3, instead of the best-5 regret registered for the v3 pools; under best-5 the patched objective’s regret (0.069) also exceeds that of the random ranking then used (0.056). The token-level re-roll study (Table 3) re-reads at the validity budget of 384 tokens rather than the registered 4,096, to match Table 2.